Kiosk Sheet Dispense Routine Logic and Information v1.0 122724

Background Information

The kiosk has a transaction where it dispenses a number of laundry sheets. Currently its set to 5 sheets per dispense transaction. We need to be able to accurately determine the number of sheets that are being dispensed. During a transaction the kiosk will start a motor (either scented or unscented motor) and then count the number of sheets dropped and then stop the motor when the number of desired sheets has been dispensed.

The sheets are counted by optical sensors (beam break sensors)

Adafruit - IR Break Beam Sensors with Premium Wire Header Ends - 3mm LEDs : ID 2167 : Adafruit Industries, Unique & fun DIY electronics and kits

We have 2 sensors per side of the kiosk so 4 total.

Scented – 1 Top sensor and 1 Bottom sensor

Unscented – 1 Top sensor and 1 Bottom sensor

The sheet will fall through the dispense chute and trip first the top sensor and then the bottom sensor. We need to use this data to count the number of sheets that have been dispensed.

Typical Expected Dispense Routine.

1. Motor starts
2. Sheet falls through top sensor – Raspberry Pi (RasPi) gets the data from the Motor Control Board (MCB)
3. Sheet falls through bottom sensor – RasPi gets the data from the MCB
4. We count this as 1 sheet.
5. We repeat until the kiosk has dispensed the number of sheets configured IE 5 sheets.

Notes: Sheets can stick together and more sheets then counted could dispense BUT that is not our responsibility. We still count this as 1 sheet.

Issues that we need to account for the in dispense routine.

1. Motor Control Board communications are not available
   Currently we have an issue with the communications between the MCB and the RasPi. Sometimes the communication is not available. We don't know why this happens but our team is looking into it.

Proposed fix – When the MCB is not communicating with the RasPi we have to use a timeout as the only method of stopping the dispense cycle. We have to assume that 5 sheets were dispensed.

Secondary Issue – If the motor control board communications are not available we do not know if there is sufficient inventory to dispense the sheets. We have to assume this and start the dispense cycle anyways. We also may possibly need to use the timeout to stop the dispense. We can check the beam break sensors after we start the dispense as sometime even though the communication is not available once we start the dispense the beam break sensor data does become available. We need to setup a testing parameter to determine this.

2. Inventory counting issue scenarios.
    - The motors that dispense sheets do not stop perfectly everytime as there are stopped just after the beam break sensors trigger them to. So sometimes when the motor is stopped the next laundry sheet to be dispensed is hanging out of the motor and is actually triggering the top bream break sensor. We need to have the code understand this anomaly.
    - The chute can become clogged so we have to have intelligent counting of the sheet drop and react to the assumption that the sheets have clogged in the chute (use the motor timeout feature)

Features we would like to implement.

1. Check the status of the limit switches to determine if there is sufficient inventory to start a dispense routine. The top limit switch on the rail if triggered would indicate that there is no more inventory available to dispense. We can use this in any routine where it would benefit.